

Project name: Cornetto: Combinatorial and Relational Network as Toolkit for Dutch Language Technology

Project number: STE05039

Reporting period: 1 April 2006 – 1 October 2007

Participants: Vrije Universiteit, Amsterdam, Faculteit der Letteren (VU-AMS)
Universiteit van Amsterdam, Instituut voor Informatica (UVA-AMS)
K.U. Leuven- ICRI (KU-LEU)
Irion Technologies B.V. (IRION)

Persons involved:

Piek Vossen (VU-AMS/IRION)
Willy Martin (VU-AMS)
Hennie van der Vliet (VU-AMS)
Isa Maks (VU-AMS)
Roxane Segers (VU-AMS)
Maarten de Rijke (UVA-AMS)
Erik Tjong Kim Sang (UVA-AMS)
Katja Hofmann (UVA-AMS)
Marie-Francine Moens (KU-LEU)
Erik Boiy (KU-LEU)
Hetty van Zutphen (IRION)
Agata Cybulska (IRION)

1 Summary

1.1 Deliverables

The next table lists the deliverables with related work packages, the date of delivery and the status. The numbering is in accordance with appendix-4 of the consortium agreement.

Table 1: Deliverables

Id	Wp	Title	delivery date	status	Mo.
	0	project management en rapportering	continuing	in progress	
D01	1	Cornetto Database	30/09/06	done	6
D02	2	Aligned RBN, DWN and Wordnet2.x	30/09/06	done	6
D03	3	Top-level ontology, relation constraints and assignments	31/12/06	delayed, in progress	9
D04	4	Core Cornetto	30/07/07	delayed, in progress	15
D05	4	Extended Cornetto	31/01/08	not yet started	22
D06	5	Acquired concepts	30/03/07	in progress	12
D07	5	Acquired relations	30/07/07	in progress	15
D08	5	Cornetto Acquisition toolkit	30/07/07	in progress	15
D09	6	Ontology for the legal and finance domain	30/03/07	delayed, in progress	12
D10	6	Sub-database tuned, trimmed and extended to the legal and finance domain	31/01/08	not yet started	22
D11	6	Domain tuning toolkit	31/01/08	delayed, in progress	22
D12	7	Evaluation report of the ontology advisory-board	30/03/07	in progress	12
D13	7	Task-based evaluation of Core Cornetto	30/09/07	in progress	17
D14	7	Task-based evaluation of Extended Cornetto	31/01/08	not yet started	22
D15	7	Evaluation report of the user-group on the legal and finance sub-database	31/01/08	not yet started	22
D16	8	Final Cornetto database and software	30/03/08	in progress	24

1.2 Finished deliverables

D01	1	Cornetto Database
D02	2	Aligned RBN, DWN and Wordnet2.x

1.3 Changes

Now the database editor is available, the editing work can start. The assignment of the ontology labels and the editing of the core database were delayed due to the late delivery of the database. No further delays are expected anymore with respect to the editing. To allocate more man power to the editing, IRION will carry out the work of the VU-AMS with respect to the ontology assignment and the manual editing. For this work, 25K of the budget of VU-AMS has been moved to IRION. This was approved by the Stevin-bureau. The work will be carried out in the next period. In this way, we hope to catch up with the delays that result from the late delivery of the database.

Due too difficulty finding a candidate at KU-LEUVEN, the domain extension was started in this period. As explained in the previous report this delay has no consequences for the other tasks in the project because there are no dependencies. We asked for an extension of the project to complete the work until 1st of July 2008. The extension in time was granted by the Stevin-bureau as far as the costs are within the limits of the budget.

1.4 Spent effort compared to planned effort

partner	Total person months whole project	Nr. person months spent for period 3	Nr. person months spent in total
VU-AMS	18	6 (7)	12(16)+
UVA-AMS	15	5.2(5.2)	13.2(17.1)+
KU-LEU	14	5 (0.2)	5 (0.33)+
IRION	10	0,5 (1)	9,5(14)+
Totaal	57		

Comment: the straight number indicate the paid effort in person months, the numbers between brackets give the actual person months including the work exceeding the paid effort.

1.5 Dissemination of the results

Papers

Period 3

Vossen, P., Hofmann, K., de Rijke, M., Tjong Kim Sang, E., and Deschacht, K. (2007). The Cornetto Database: Architecture and User-Scenarios. In DIR 2007, pp. 89-96. ([PDF](#)) Publically available on external Website

Vossen P. and C. Fellbaum, 2007 (fc) "Universals and Idiosyncracies in Multilingual WordNets", In: Handbook Multilingual Lexicography, [Oxford University Press](#), 2007

Period 2

Fellbaum C. and P. Vossen, 2007 "Connecting the Universal to the Specific: Towards the Global Grid", In: Proceedings of [The First International Workshop on Intercultural Collaboration](#) (IWIC 2007), Kyoto, Japan, January 25-26, 2007 Publically available on external Website

Vossen, P. (2006). *Cornetto: Een lexicaal-semantische database voor taaltechnologie*, Dixit Special Issue, Stevin. ([PDF](#)) Publically available on external Website

Vossen, P. (2006). *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. ([PDF](#)) Publically available on external Website

Posters

Period 3

Tjong Kim Sang, E. (2007) Extracting Hypernym Pairs from the Web. In Proceedings of ACL 2007 (poster), Prague, Czech Republic.

Period 2

Hofmann, K. and Tjong Kim Sang, E. (2007). Automatic Extension of Non-English WordNets. SigIR 2007, accepted. Publically available on external Website

Talks

Period 3

Tjong Kim Sang, E.: Cornetto presentation on the [Stevin](#) Programmadag, Nederlandse TaalUnie, Hoeven, September 21, Publically available on external Website

Vossen, P, Guest lecture: "Cornetto: Een Combinatorische Semantische Database voor het Nederlands" for the [Capita Selecta](#) of the Instituut voor Nederlandse Lexicologie ([INL](#)), Leiden, May 22, 2007. Publically available on external Website

Vossen, P, Guest lecture: Global WordNet and Global WordNet Grid at the [University of Potchefstroom](#), South Africa, March 19, 2007. Publically available on external Website

Vossen, P, Guest lecture: Cornetto and Wordnets at the [University of Potchefstroom](#), South Africa, March 20, 2007. Publically available on external Website

Vossen, P, Guest lecture: "African Languages WordNet Workshop/Training" of the [Department African Languages](#) of the University of South Africa, Unisa), Pretoria, South Africa, March 21-23, 2007.

Period 2

Tjong Kim Sang, E.: "Extracting Dutch Hypernymy Pairs from the Web". Presented at CLIN-17, 12 January 2007 in Leuven, Belgium. Publically available on external Website

Vossen, P., Maks, I., Martin, W., van der Vliet, H., Hofmann, K., van Zutphen, H.: *The Cornetto Database*. Presented at CLIN-17, 12 January 2007 in Leuven. ([Abstract](#), [PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. Presented December 22 2006 at Vrije Universiteit, Amsterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Er kan meer dan men doet met de Cornetto database*. Presented November 30 2006 at the TST Themadag: de gebruiker centraal in Rotterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Period 1

Vossen, P.: *Cornetto*. Presented September 11 2006 at the Stevin project day in Antwerp. ([PDF](#), [PowerPoint](#)) Publically available on external Website

2 Progress per deliverable

2.1 Realized work in the reporting period

WP0: Management

In the 3rd period, we had 4 project meetings:

- June 21st, 2007, Universiteit van Amsterdam
- June, 7 & 8, Vrije Universiteit Amsterdam (Workshop)
- August, 23rd, 2007, Universiteit van Amsterdam
- September, 21st 2007, Stevin programmadag, Hoeven

In addition, we had technical meetings between the individual partners of the project.

We organized a project workshop on the 7th & 8th of June, where we invited the members of the user-group and the ontology-group for a discussion on the results so far. In addition to the user-group, we invited people and research groups that have expressed their interest in using the database. In total, about 30 people attended the workshop. We invited Adam Pease to give an invited talk about SUMO. On the 8th of June, we had a project meeting on the ontology assignment in Cornetto with Adam Pease. In this meeting we discussed our proposed model and we finalized our approach.

WP1: Cornetto database

The structure of the database and the editors were adapted to store ontology mappings. Some measures have been carried out to improve the speed of the database. Various bug fixes have been made. This work package has been completed.

WP2: Aligning semantic resources

We carried out a separate alignment of the Dutch WordNet with the English Wordnet 1.6, based on the heuristics developed in the MEANING project. These new mappings have been imported in the Cornetto database and have been used to import SUMO and WordnetDomain labels into the database. This work package has been completed.

WP3: Ontologize

We defined the mapping relations between the ontology and the wordnet database. This is published and presented as part of the Global Wordnet Grid project (see publications and presentations). This model makes a clear distinction between the ontology and the synsets and allows for explicit mappings between synsets and the ontology through a naming relation, a subsumption relation or through complex expressions. These complex mapping relations are based on lists of triplets containing variables, SUMO terms and SUMO relations. The XSD of the database was extended to store the triplets.

We created an improved mapping between the Dutch wordnet synsets and the English wordnet synsets. Through this more precise mapping, we could create an import that stores the equivalence mappings, the Wordnet domain labels and the SUMO mappings from the English Wordnet to the Dutch synsets. This import has been the starting point for the assignment of the SUMO labels to the Dutch wordnet.

The mappings have been revised for a number of critical cases. Furthermore, we worked out a number of complex mappings. These were presented and discussed on the workshop.

WP4: Cornetto editing

The editing of the aligned RBN-DWN was carried out in 5 different groups of words:

Editing and alignment RBN-DWN:

Adjectives (frequent and / or polysemous)	50	words
Adjectives with 2 senses :	110	
Frequent verbs:	130	
Frequent nouns:	80	
Nouns with a focus on semantic type shifts	550	
Total	920	

The total of 920 words correspond with about 4000 lexical units and synsets.

Other work involved compiling guidelines for editing and aligning adjectives (with a focus on emotion-related adjectives), nouns (with a focus in semantic type shifts), and verbs.

WP5: Acquisition toolkit

Our work on deriving lexical information automatically (see papers by Tjong Kim Sang and by Hofmann) have shown that while current techniques are able to retrieve correct relation items from text, the attained levels of precision of the methods are not good enough to allow automatic inclusion of results into the database.

Rather than shifting the burden of manually checking noisy and perhaps irrelevant results on the lexicographers, we have chosen a different approach. We are making the acquisition toolkit available to the lexicographers, allowing them to search for information that is relevant to their current information needs, thereby assuring that the data produced by the toolkit is relevant, if it is correct.

At present we have offered an online toolkit with two query methods to the lexicographers. We are processing the feedback received on the toolkit.

WP6: Domain acquisition

We collected domain specific corpora and carried out a literature study. Furthermore, we made a planning for developing the domain acquisition tool. Preliminary test for detecting domain specific terms in the legal-financial corpus have been done.

UvA has started processing domain-specific corpus material from Leuven and will provide access to a command line version of the toolkit to Leuven.

WP7: Evaluation

A plan for an extrinsic evaluation the Cornetto database has been developed in the previous time frame. UvA and IRION are now waiting for the first stable version of Cornetto in order to start the evaluation.

WP8: Distribution

University of Amsterdam has signed the license agreement as well. The contract is now sent to the Taalunie for final signature.

2.2 Problems and solutions

- No further problems

2.3 Suggested planning for the next period

WP3: Ontologize

The ontology assignment that is imported from the English wordnet is further revised and improved for important concepts. This will involve:

- frequent and ambiguous words
- synsets with many equivalent mappings to English Wordnet
- concepts with high position in the hierarchy
- concepts with conflicting ontology assignments

When the manual work is done for the frequent & polysemous words and for the top concepts, we will extract all inherited SUMO labels for all synsets and check these for conflicts. The conflicts will be traced to their cause and corrected.

WP4: Cornetto editing

Continue editing of problematic cases:

- **Editing and alignment RBN-DWN:**

Adjectives (frequent and / or polysemous)	250	words
Frequent verbs	170	
Frequent nouns	170	
Nouns with a focus on semantic type shifts	600	
Polysemous nouns (6 or more meanings)	514	
	1454	

The editing is extended to complete the SUMO mapping, the equivalence mapping to English wordnet and the domain assignment. This asset corresponds with another 5000-6000 lexical units and synsets.

We further focus the editing in the last phase on:

- concepts with many descendants (hyponymy and meronymy);
- concepts with many unmatched LUs and synsets;
- concepts with many equivalence mappings to English wordnet;
- concepts with conflicting SUMO labels and Domain labels;

Finally, we will make sure that all manually edited LUs and synsets will be part of the complete hierarchy in the sense that their hyperonyms are also manually verified. In this way we guarantee a semantic closure of part of the database.

- **Linking DWN-SUMO**

Linking of all aligned synsets (approx. **6000**) to SUMO ontology

- **Compiling guidelines** for linking adjectives, nouns, verbs to the SUMO ontology

MultiWordUnits

- Compiling guidelines for selection, editing, and alignment of MultiWordUnits
- Aligning to DWN, linking to SUMO of a set of 100 MultiWordUnits

WP5: Acquisition toolkit

UvA proceeds and finishes the development of the online toolkit and makes a small amount of automatically derived concept and relation information to the lexicographers for evaluation purposes.

WP6: Domain acquisition

In the next period, we will concentrate on the recognition of domain specific words, including compounds. We will carry out an extensive evaluation. We will also start with the design of software for detecting word meanings and collocation of domain specific words.

UvA provides command line version of toolkit with preprocessed data to Leuven and offers support in their initial experiments with the toolkit.

WP7: Evaluation

UvA performs an extrinsic evaluation of a version of the Cornetto database.

Irion will implement different methods to implement semantic relations in the classification software. One method is based on adding any content word combination to the index and for the classification and the second method only combines combinations that are detected using the Cornetto database. We will investigate the effects compared to a baseline using the standard evaluation package that is part of the classification software.

WP8: **Distribution**

The final license agreements will become available. We will set up a second server at VU-AMS so that users can access the database without hampering the editing.